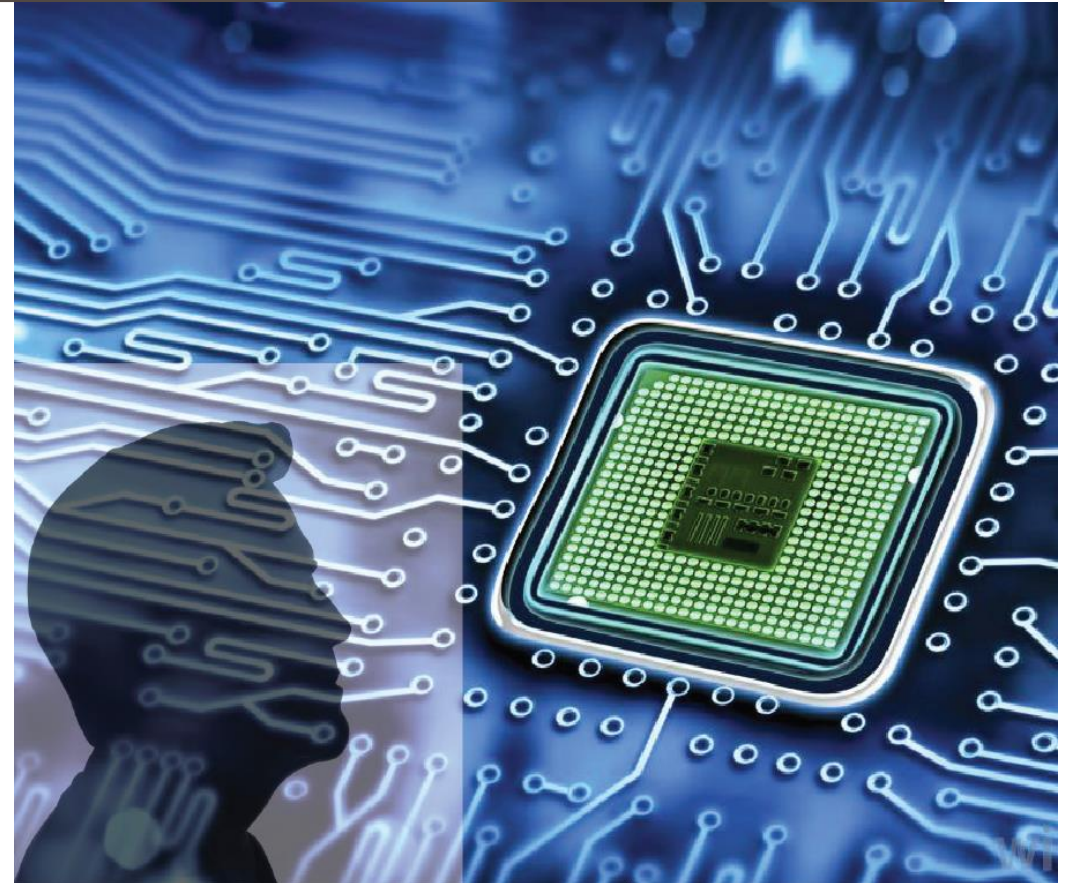


VLSI DESIGN FLOW

Week-9 Lecture-1 STA

Sneh Saurabh
14th October, 2024



VDF: Revision/Announcements

-



Static Timing Analysis

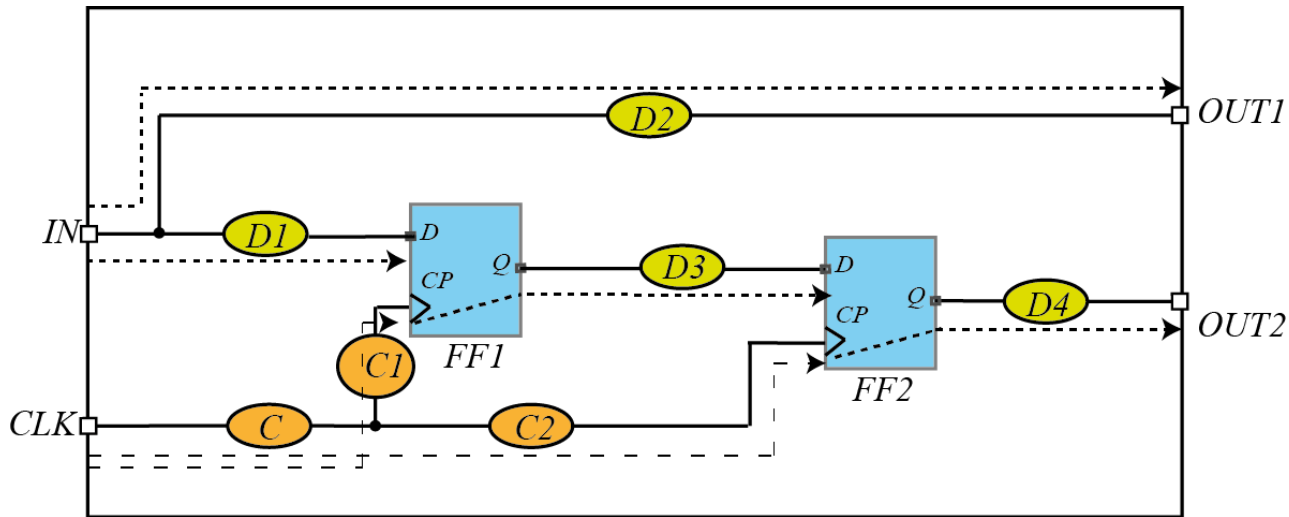


Basics

Static Timing Analysis: Path Types

STA considers two types of paths:

1. Data Path
2. Clock Path



- Four data paths

- Two clock paths

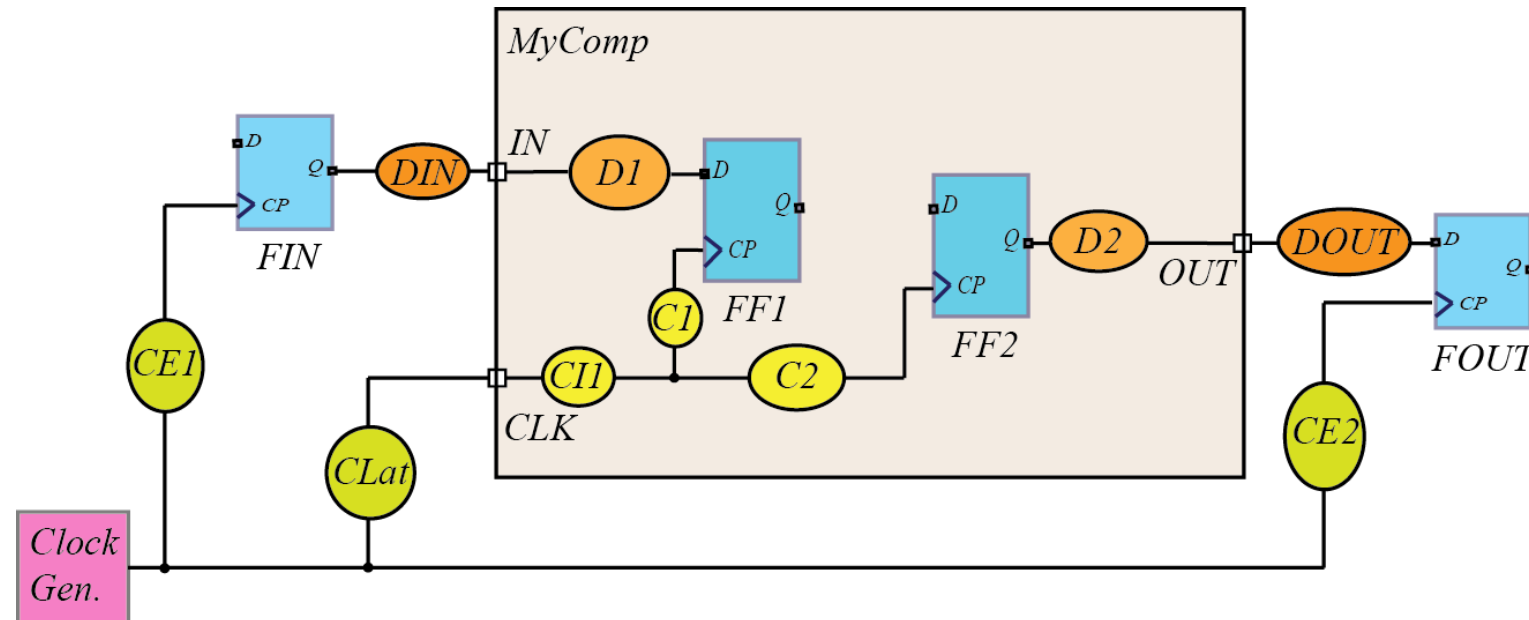
Data Path:

- **Timing startpoints:** input ports or clock-pin of a flip-flop
- Goes through combinational circuit elements
- **Timing endpoints:** D-pin of a flip-flop or output ports
 - setup and hold checks are performed at the timing endpoints

Clock Path:

- Starts at clock source (specified in *constraints*)
- Passes through the combinational circuit elements (buffers and inverters)
- Ends at a clock-pin of a flip-flop

Static Timing Analysis: Environment of design



- The environment of design puts additional constraints on the signal entering/leaving a design
- Signal entering the design at the input port *IN* has delay due to logic represented by *CE1*, *FIN*, and *DIN*
- Signal leaving the design at the output port *OUT* will encounter further delay due to logic represented by *DOUT* and should meet setup/hold requirements at *FOUT*
 - ⇒ Check for setup/hold must also be performed at output port

Constraints

Constraints: Basics

Constraints are:

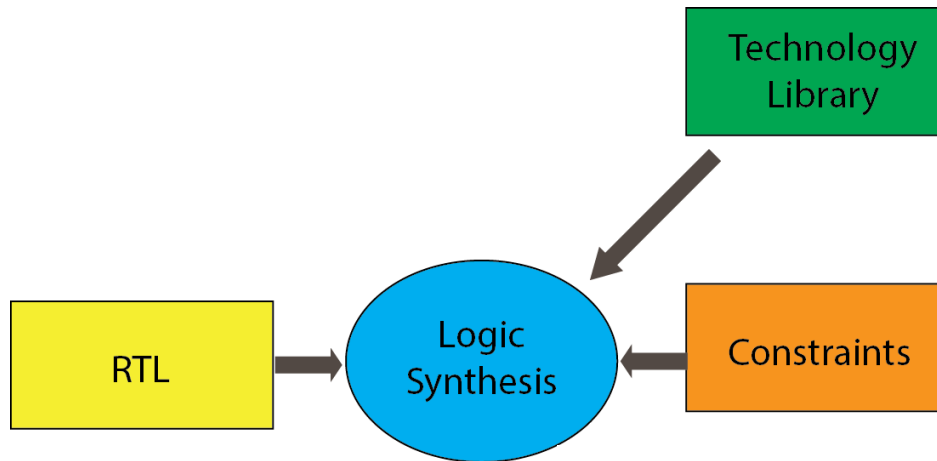
- **Requirements** of a design that needs to be honoured or attempted to be honoured by the CAD tools
- **Information** about a design that could potentially be exploited by the CAD tools to improve the PPA of the design

- Constraints are normally specified in **Synopsys Design Constraints (SDC)**
- ASCII format written in **Tool Command Language (TCL)**

```
create_clock -period 10 -waveform {5 10} [get_ports CLK]  
set_clock_transition -rise 0.1 [get_clocks CLK]  
set_clock_uncertainty 0.2 [get_clocks CLK]  
set_input_delay -clock CLK 3.0 [get_ports INPA]  
set_output_delay -clock CLK 3.0 [get_ports INPA]  
set_false_path -from [get_ports TE]  
set_multicycle_path -from [get_ports mult_out]
```

Constraints: Application

- Most of the constraints are related to *timing* of a design (also called timing constraints)
 - Employed by implementation tool to gather information about the expected timing behavior
 - Employed by STA tool to verify timing



Constraints: What is the origin?

Constraints are normally manually written

- Designer have the knowledge of design goals and environmental constraints
- There are certain tools for automatic constraints generation
 - Some user intervention is always required

It is important to write correct constraints

- Otherwise design implementation tools can produce unexpected results
- There should be consistency between different constraints
- Constraints should be consistent with design attributes

Constraints: Types

Clock signal

- Attributes of a clock signal such as frequency, duty cycle, skew, uncertainty and delay

Environment under which the design operates

- Attributes of external incoming signal and expected behavior of the signals produced by a design

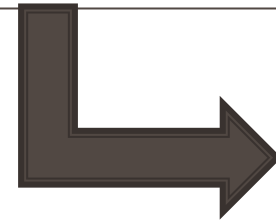
Functionality of the design (informative)

- Timing exceptions (paths that are *false* and paths that are allowed to behave differently than traditional synchronous behavior), modes of design

Design rules and optimization constraints

- Maximum slew at the port, maximum capacitance at a pin, and soft constraints

Constraints

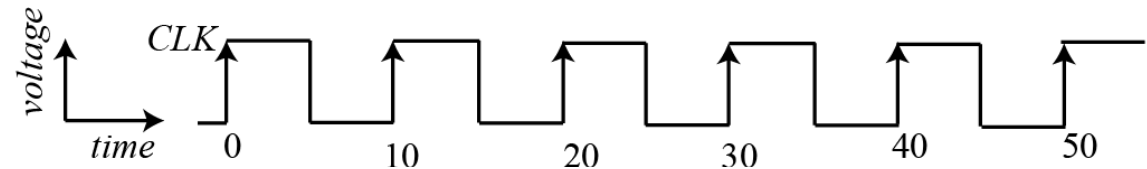
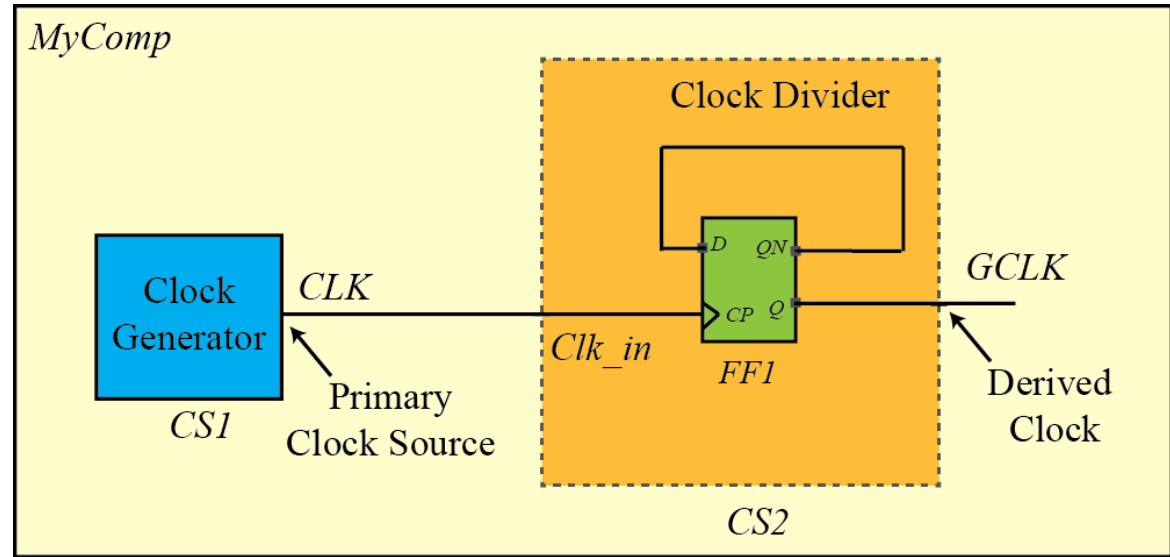


Clock Signal

Clock Constraints: Sources

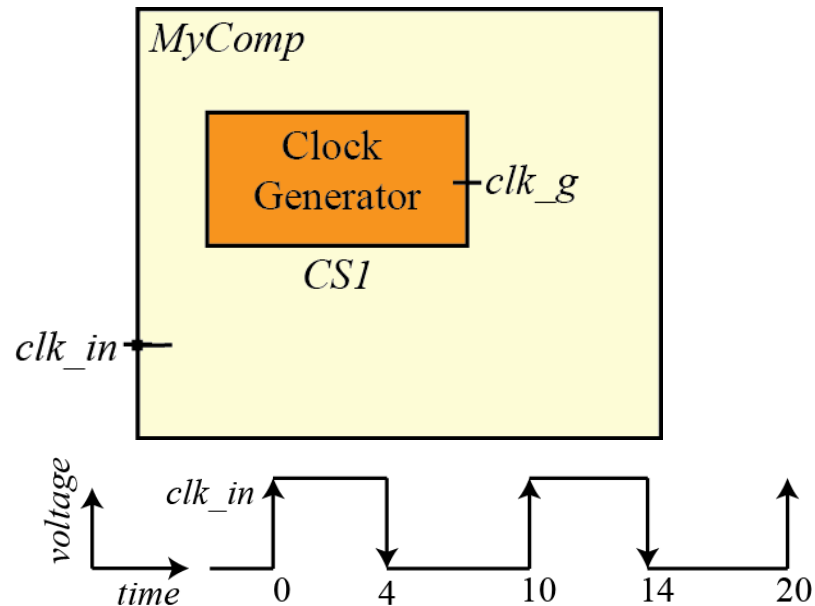
Two types of clock sources

- **Primary clock sources:** waveform independent of other clock sources in that design
 - **Derived clock sources:** waveform depends on other clock sources
-
- **Master clock:** clock from which we derive another clock is known as the master clock of the derived clock
 - *CS1* is the master clock source of *CS2*



Primary Clock Source Definition

create_clock: defines the primary clock source in a design

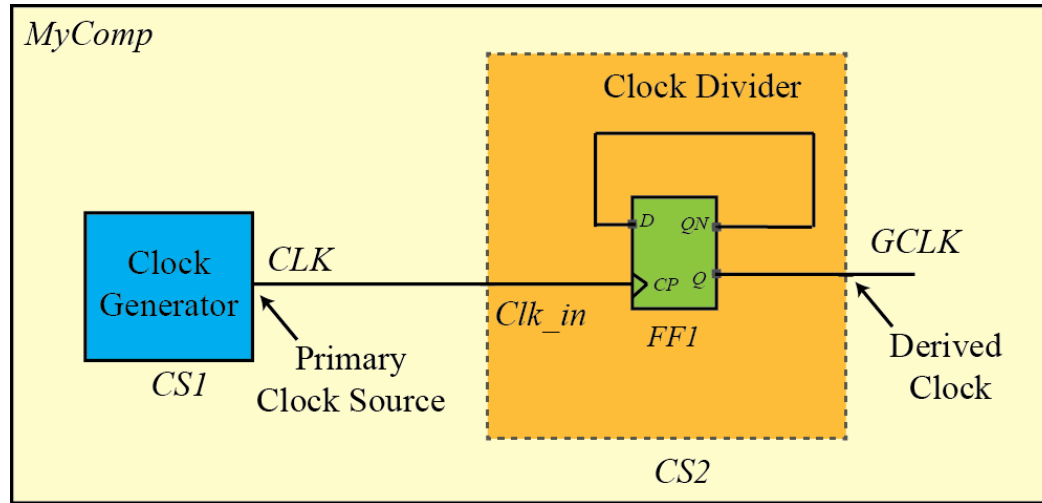


```
current_design MyComp
create_clock -name EXT_CLK -period 10 -waveform {0 4}
[get_ports clk_in]
create_clock -name INT_CLK -period 10 [get_pins CS1/clk_g]
```

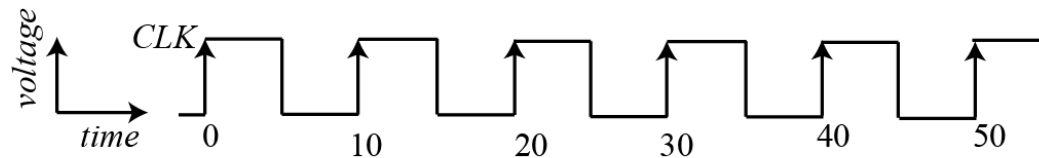
-waveform: time when the clock edges occur, starting from rise-edge

Derived Clock Source Definition

create_generated_clock: define derived clock sources

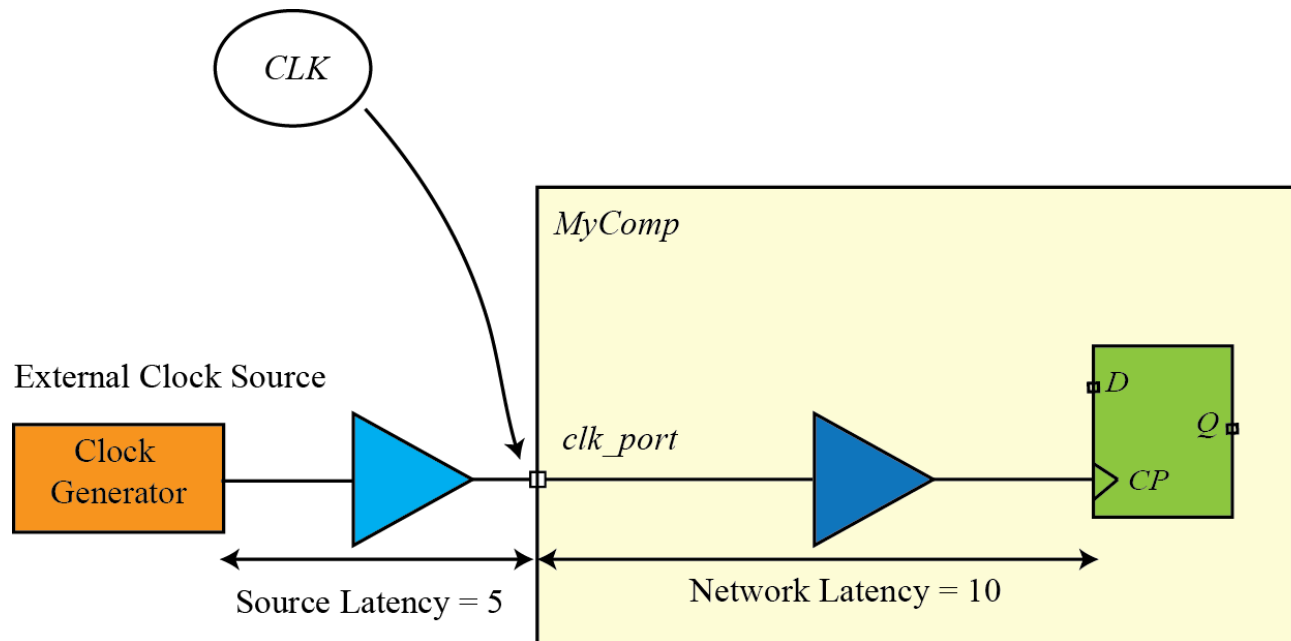


```
create_clock -name CLK -period 10 [get_pins CS1/CLK]
create_generated_clock -name GCLK -divide_by 2
-source [get_pins CS1/CLK] [get_pins CS2/GCLK]
```



Attributes of Clock Signal: Latency

set_clock_latency: specify clock latency



```
create_clock -name CLK -period 200 [get_ports clk_port]
set_clock_latency 5 -source [get_clocks CLK]
set_clock_latency 10 [get_clocks CLK]
```

Attributes of Clock Signal: Uncertainty

set_clock_uncertainty: unpredictable deviation of the clock edges from the ideal value

Clock uncertainty can be used to model:

- Jitter: temporal variation
- Skew: spatial variation
- Safety margins

```
create_clock -name CLK -period 200 [get_ports clk_port]  
set_clock_uncertainty 15 -hold [get_clocks CLK]  
set_clock_uncertainty 20 -setup [get_clocks CLK]
```


References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.